

2

Modulation of Power Electronic Converters

The power electronic energy converter operates as a power amplifier in an electrical control system. The energy conversion of the process can be influenced by control of the voltage fed to the load. The converter is capable to convert one voltage type to another, i.e. from dc to ac. The conversion can also be from one dc level to another or from one ac voltage to another.

If the converted power is low, a continuous control of the output voltage can be used, just like in an audio amplifier to a loudspeaker. The transistors operate then in their linear mode and significant parts of the converted power become losses. Such a control method needs large components and heat sinks already at small converted powers.

At higher power ratings the output power is controlled by modulation and the transistors are used as switches. The switch valves (power transistors, GTO-thyristors etc.) are either short circuit or open circuit. Due to the low voltage drop when conducting, typical 1-3 volts, and the short switching periods, typical $<1 \mu\text{s}$, the losses relative to the converted power is considerably reduced. The power density of the converter, i.e. the converted power divided with weight or volume, is in this way radically increased.

There are several drawbacks with the modulation of the output voltage. The large harmonic content results in a certain current ripple in the load current. A common consequence of this is radio interference and acoustic noise and vibrations. Torque pulsations and increased losses are the results in electrical machines. Therefore it is important to develop methods to decrease these consequences.

In a complete control system usually the converter and it's control are the lesser part. Therefore it is important to achieve a good co-operation with other parts of the system that the power converter acts like a module with the help of suitable interfaces. At commissioning, every module can be tested independently. One such interface is to design the converter and it's control to act like a voltage or current generator.

This chapter describes the modulation of several different power converters to obtain a desired output voltage. The modulation methods described are of carrier wave type, whereas other types of modulation that involve direct control of a secondary quantity like current, torque or magnetic flux linkage are describe in chapter 4. The goal with the modulation is to obtain linearity between the reference of the output voltage and the average of the output voltage, as measured over one carrier wave period.

2.1 The converter topology

Every converter is built according to the functionality of the power semiconductors. A power semiconductor (= valve) should not be used to connect two strong voltages, i.e. two capacitors with different charge, since the current transient through the valve would be very high. The valve should neither be used to connect two inductive circuits with different current, since the voltage transient would be extremely high. Every valve must be used to connect one voltage stiff circuit with one current stiff. Power converters are usually built with one capacitive side and one inductive. In Figure 2.1 this is shown for four different types of power converters. They have been drawn with the voltage stiff side (capacitive) to the left and the current stiff side (inductive) to the right.

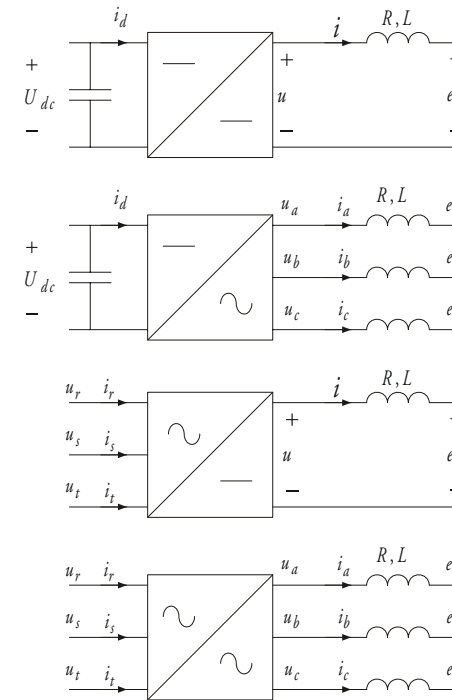


Figure 2.1: Different converter types.

The voltage on the inductive side of the converters can be modulated, since the inductance gives high impedance to the voltage transients and balances the voltage difference between the converter output voltage and the load voltage. The voltage ripple is due to the output voltage ripple whereas e varies with a much lower frequency than the modulation frequency. The variation of e is assumed to be negligible during one modulation period.

Correspondingly, the current can be modulated on the voltage stiff side (capacitive) since that side has low impedance to the current pulses. Self-commutated converters have distinct inductive and capacitive sides whereas grid or load commutated converters usually have a small commutation inductance on the voltage stiff side as well. This inductance is usually neglected in the first approximation.

Since the voltage modulation means that the voltage pulsates, the power flow will also pulsate. The converters discussed here are assumed to be ideal, and they do not contain any energy storing components such as capacitors and inductances. The momentary power will therefore be equal on the primary and secondary side. The inductances on the current stiff side and the capacitances on the voltage stiff side are used to even out the power pulsations between the source and the load.

Different types of converters have in common that the voltage can be modulated on the inductive side. Voltage modulation means, that the momentary value of the output voltage alternates between well defined levels, i.e. two dc voltage levels or voltage curves defined by the grid voltage. The momentary value of the voltage is thus not controllable in every time instant. On the other hand, since the output voltage is modulated periodically, the average value and the voltage-time-area can be controlled for every modulation period.

The output voltage from grid- or load-commutated converters has a saw tooth like shape with a constant commutation frequency being a multiple of the line frequency. For the most common three phase connection, the modulation frequency is 300 Hz. Self commutated converters can select the modulation frequency more freely and it doesn't have to be constant. For transistor converters it is possible to have modulation frequencies in the 10-100 kHz range, whilst large GTO converters limits the modulation frequency to about 1000 Hz.

To arrange a unified approach, valid for both line commutated and self commutated converters, this description will be limited to modulation with constant frequency. The modulated voltage has the period time T and the control is aimed at the voltage time area y in every modulation period. The relation between the voltage time area and the average voltage during the switch interval is:

$$\bar{u} = \frac{y}{T} \quad (2.1)$$

The voltage time area is controlled after a reference y^* , and the goal is that the real area and the reference should be equal.

$$y = y^* \quad (2.2)$$

In the next section of this chapter, a description of the control of the voltage time area is given.

2.2 Modulation. Control of voltage time area

The most important principles for modulation can be illustrated with the circuit in Figure 2.2. A two-position switch controls the potential in c . When the switch switches periodically, the output voltage is modulated periodically.

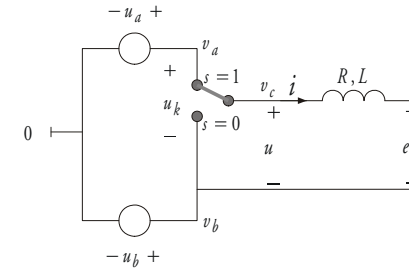


Figure 2.2: Circuit with switch to give pulse modulated output voltage. *)

The potential v_c can alter between the potentials v_a and v_b and the relation with the switch position s is

$$v_c = \begin{cases} v_a & \text{when } s = 1 \\ v_b & \text{when } s = 0 \end{cases} \quad (2.3)$$

The output voltage u , connected to a load with a resistance (R), inductance (L) and the counter emf (e) in series, is (pulse-) modulated as a function of the switch position

$$u = s \cdot (v_a - v_b) = s \cdot u_k = \begin{cases} u_k & \text{when } s = 1 \\ 0 & \text{when } s = 0 \end{cases} \quad (2.4)$$

The voltage u_k is the voltage step achieved when the switch is moved from one position to another. In this context the step is assumed to be equal to the difference between a and b .

Figure 2.3 gives an example of modulation of output voltage. The potential v_c alternates between the levels v_a and v_b whilst u alternates between 0 and u_k . A modulated voltage like the one in Figure 2.3 can be controlled with the ratio of the pulse width in relation to the pulse gaps. In addition, the supply voltage influences the surfaces.

*) Note. 1. This circuit doesn't describe the case with discontinuous current.

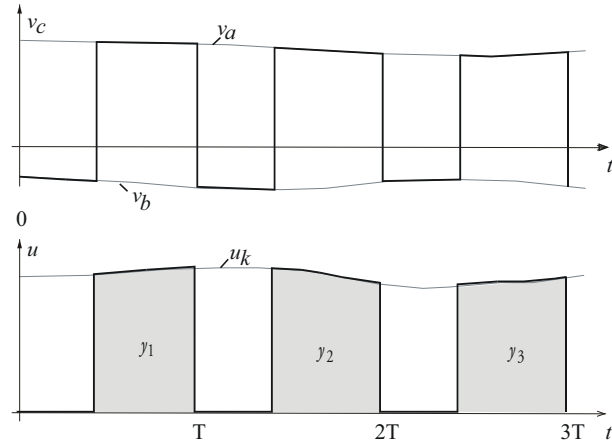


Figure 2.3: The modulation of the potentials v_a and v_b and the modulation of the output voltage between 0 and u_k .

When the modulation is limited within a period T , and a slowly varying voltage u_k , there are three fundamental ways to influence the surface

1. Variation of the position of the positive flank
2. " " " " negative "
3. " " both the negative and the positive flank.

These three cases are shown in Figure 2.4. The maximum Y_0 that can be achieved during the period T is according to Figure 2.4 a.

$$Y_0 = \int_0^T u_k(t) \cdot dt \quad (2.5)$$

To control the area the position τ_+ for the positive flank and τ_- for the negative flank is varied. The relation between a small variation $d\tau_+$ or $d\tau_-$ and a small change in the output area dy , can be written

$$\begin{aligned} u_k(\tau_+) &= -\frac{dy(\tau_+)}{d\tau_+} \\ u_k(\tau_-) &= \frac{dy(\tau_-)}{d\tau_-} \end{aligned} \quad (2.6)$$

With control of the positive flank, see Figure 2.4 b, the output surface is,

$$y(\tau_+) = \int_{\tau_+}^T u_k \cdot dt = Y_0 - \int_0^{\tau_+} u_k \cdot dt \quad (2.7)$$

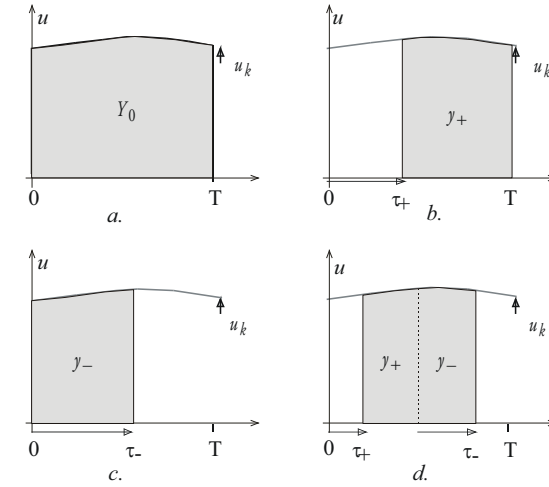


Figure 2.4: Control of voltage time area. a) Maximum area, b) variation with positive flank, c) variation of negative flank and d) variation with both flanks.

This equation describes how the voltage time area is reduced more the longer the positive flank is delayed. If Y_0 is known during the period, the surface $y(\tau_+)$ is determined already at τ_+ i.e. before the voltage pulse really occurs. This is an important property that later on will be used for control of the surface.

Control with only the negative flank, see Figure 2.4 c, gives the surface

$$y(\tau_-) = \int_0^{\tau_-} u_k \cdot dt \quad (2.8)$$

The surface grows in this case with τ_- and when the negative flank comes, the output surface is ended. In the last case, see Figure 2.4 d, both parts of the surface are controlled separately

$$y(\tau_+, \tau_-) = y_+ + y_- \quad (2.9)$$

Every part of the surface can be written as

$$y_+ = \int_{\tau_+}^{T/2} u_k \cdot dt = Y_0 - \int_0^{\tau_+} u_k \cdot dt \quad (2.10)$$

and

$$y_- = \int_{T/2}^{T/2+\tau_-} u_k \cdot dt \quad (2.11)$$

where Y_0 in this case is

$$Y_0 = \int_0^{T/2} u_k \cdot dt \quad (2.12)$$

The surfaces under a given voltage u_k in a modulation period T can thus be described with the variables τ_+ and τ_- according to the equations (2.5) to (2.12). When controlling the output surface, the logics are the opposite. From a given reference for the expected voltage time area (y^*) the switch times τ_+ and τ_- have to be determined. Using the expressions for the voltage time areas above can do this translation. Note that τ_+ and τ_- have to be determined "on-line" which is possible, since both variables implies waiting for the right time for the flank, i.e. for the switch to change position. The method for doing this calculation is developed in the following chapters.

2.3 Carrier wave modulation

A reference value y^* for the desired average voltage over one switching period is calculated by an external control system (current controller, see chapter 4). The pulse width must be controlled to make the voltage time area equal to the reference value. One method is to use a modulating wave y_m that is compared with the reference y^* .

To accomplish the desired result, the carrier wave has to estimate the surface as a function of the position of the flank. If the time function of the carrier wave within the modulation period is

$$y(\tau) = y_m(\tau) \quad (2.13)$$

where $y(\tau)$ is determined by the equations (2.5) to (2.12), it is also a signal that for every time instant τ has a value that, if the switching takes place in that time instant, corresponds to the resulting surface. By detecting when the carrier wave and the reference value are equal, and by at that time instant flip the switch, the problem is solved. There is however difficulties related to the generation of a carrier wave that estimates the real surface. Y_0 has to be known in the beginning of the interval. For the three cases shown in Figure 2.4, the corresponding carrier waves are given in Figure 2.5.

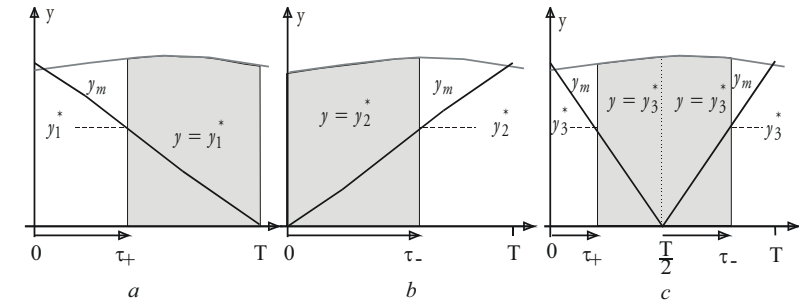


Figure 2.5: Carrier wave for three different types of modulation.

When the carrier wave and the reference are equal, the switch is switched. This is repeated every modulation period. Note that the amplitude scale in Figure 2.5 c is different from the ones in figure Figure 2.5 a,b, since the maximum voltage time area (Y_0) is calculated based on half the interval length as compared to the two cases where only the position of one flank is controlled.

The methodology with carrier wave modulation, which is focused on in this book, is fundamental for the control of several types of converters. It is however not the only method possible. With μ -computers, other methods based on prediction and other calculations are possible. It will however be proven in chapter 4 that when used properly carrier wave modulation in most cases is the best method for modulation of power electronic converters.

There are also methods where voltage modulation and current control cannot be separated. This type of control is often called tolerance band control of flux, current or torque. When torque and flux are controlled that way, the method is referred to as "DTC"-Direct Torque Control. Such methods are described in detail in Chapter 3.

Voltage time area vs. average voltage in the modulation

This far the modulation has been described with voltage time areas, both regarding the estimate of the output voltage time area as a function of the switching time instant, i.e. the modulating wave y_m , and the references for the output voltage time areas y^* .

Voltage time area as an entity is very straightforward since it emphasizes that the shape of the output voltage is not relevant, only the average value. Of course this can also be expressed as average voltages rather than voltage time areas. Experience of teaching this subject for a decade has shown that students generally prefer the average voltage, probably since voltage is easier to comprehend (can be measured with any hand held universal instrument) than voltage time area.

Thus, in the following sections and chapters, voltage time area is replaced with average voltages. No particular notation is used; the assumption of average values is implicitly applied to modulating waves and reference values according to

$$u_m = \frac{y_m}{T_s} \quad (2.14)$$

$$u^* = \frac{y^*}{T_s}$$

The implicit assumption of average values is only valid for modulating waves and reference values according to (2.14), whereas output voltages and potentials are instantaneous unless otherwise stated.

The modulator

The method with a carrier wave u_m that together with the reference value u^* determines the position of the switch, is implemented in a unit called *the modulator*. Based on the previous description, a method for modulation will be developed.

The modulator consists of a unit to compare the carrier wave with the reference value. This can either be done in an analogue modulator with a triangular wave generator and comparators, or in digital modulator using a timer circuit or an up/down counter with a digital comparator.

The carrier wave has the period time T and the amplitude depends on Y_0 , which is proportional to u_k . The carrier wave amplitude will vary as a function of u_k , something that also the amplitude of the voltage pulse does. For a given (e.g. constant) reference value for the average output voltage, small (and slow) variations in the dc link voltage will not affect the real average output voltage.

PWM-controlled dc converters

In Figure 2.6 four different dc converters are shown. In the right column of the figure, the corresponding equivalent circuit is shown. The modulation method use is called Pulse Width Modulation (PWM). The dc link voltage varies slowly in the interval $[0, U_{dmax}]$.

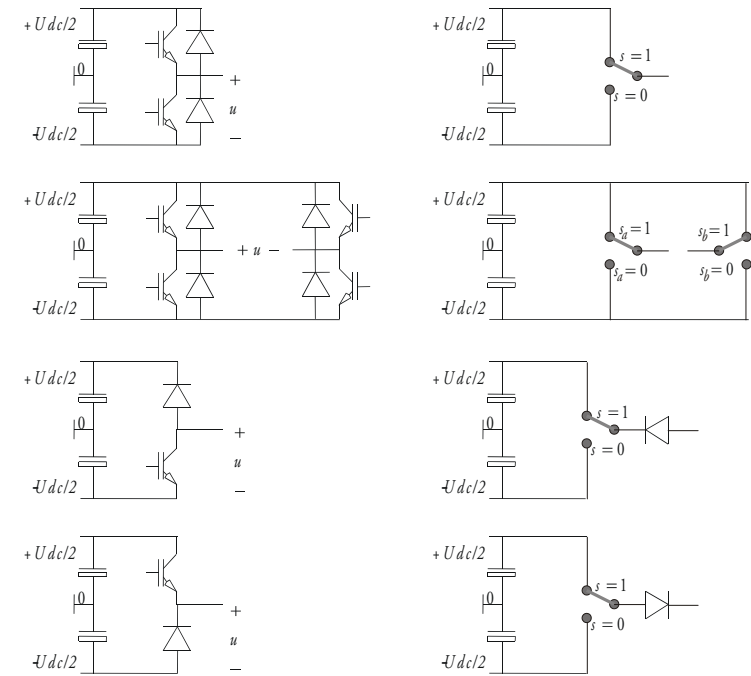


Figure 2.6: DC converters. From top: 2-quadrant, 4-quadrant, 1-quadrant step up and 1-quadrant step down

2.4 Two quadrant DC converters and One quadrant DC converters with continuous current

The current can flow in both directions in a 2-quadrant DC converter (Figure 2.7) whereas the current has only one possible direction in a 1-quadrant DC converter. As long as the current is continuous in a 1-quadrant DC converter, it works in the same way as a 2-quadrant converter with respect to the modulation.

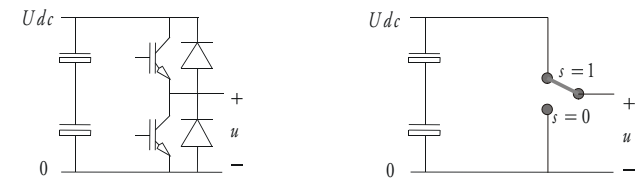


Figure 2.7: 2Q Converter. The switch equivalent is also relevant for the 1Q converter if the current is continuous.

In the first case, with control of only the positive voltage flank, the carrier wave becomes a saw tooth wave according to

$$u_m(\tau_+) = U_{dc} \cdot \left(1 - \frac{\tau_+}{T}\right) \quad (2.15)$$

The principle for the modulation is displayed in Figure 2.8. A signal generator generates a carrier wave with the amplitude U_{dc} and the desired frequency.

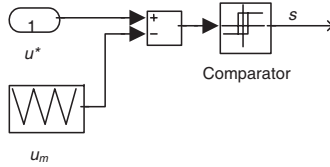


Figure 2.8: Generic structure of a modulator. Note the use of Simulink® symbols.

One example of pulse width modulation with this type of carrier wave is shown in Figure 2.9 below. The value of the reference signal in the cross over instant will determine the whole voltage time area achieved in that certain interval.

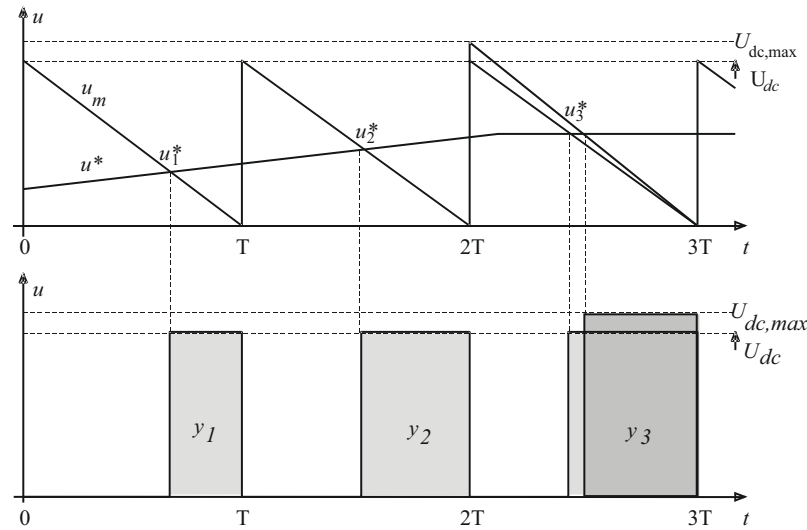


Figure 2.9: Example of control of the position of the positive flank

The maximum value of the dc link voltage and corresponding values of the carrier wave are also shown in the figure. Since the carrier wave amplitude is proportional to the dc link

voltage, the resulting voltage time area is independent of the dc link voltage. An increase in the dc link voltage, e.g., results in a higher but shorter voltage pulse with maintained area.

If instead the position of the negative flank is controlled, the saw tooth wave will change shape slightly according to

$$u_m(\tau_-) = U_{dc} \cdot \frac{\tau_-}{T} \quad (2.16)$$

i.e. a saw tooth with positive slope. The whole modulation in all other aspects equal to the case where the positive slope is controlled.

The third possibility is to control both the positive and the negative flanks. In this case the carrier wave becomes triangular. According to the equation below it is built up of two parts:

$$u_m(\tau_+, \tau_-) = \begin{cases} u_{dc} \left(1 - \frac{\tau_+}{T/2}\right) \\ u_{dc} \frac{\tau_-}{T/2} \end{cases} \quad 0 < \tau_+, \tau_- < T/2 \quad (2.17)$$

With this method, the voltage pulse can be influenced two times every switching period according to the example in Figure 2.10.

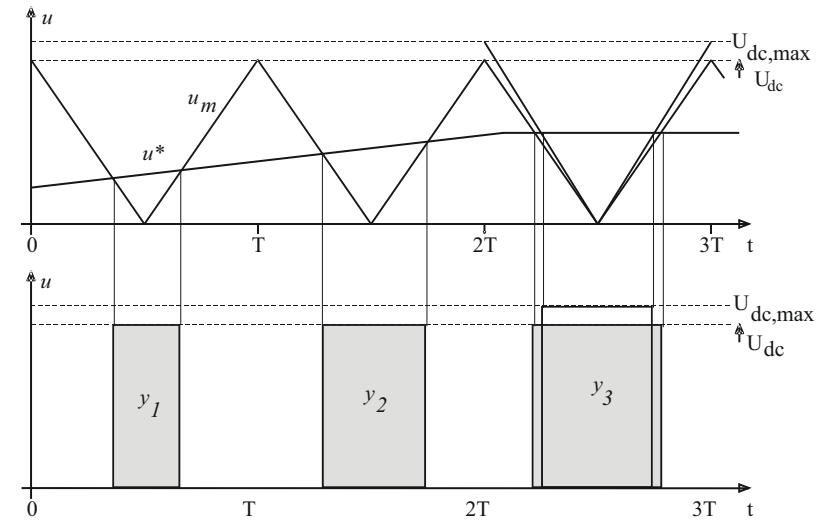


Figure 2.10: Example of control by variation of the position of both flanks.

The differences between these three methods might seem small, but they will turn out to be very important when it comes to current control. Furthermore the cases with discontinuous current with the 1-quadrant connections must be treated separately. It will be shown that the same carrier wave with small changes can be used even for 4-quadrant converters and 3 phase converters.

2.5 Four quadrant (bridge connected) converters

In the bridge connected DC converter (Figure 2.11), each branch (a and b) is controlled independently. The mid point of the DC link voltage is used as reference for the calculation of voltage time area.

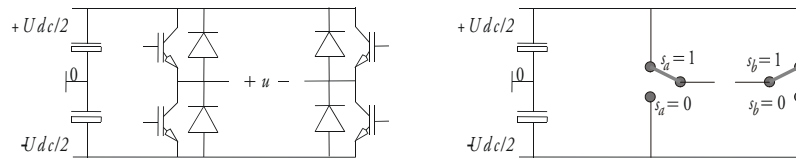


Figure 2.11: 4Q converter.

The phase potentials v_a and v_b are modulated between $\pm U_{dc}/2$, see Figure 2.12.

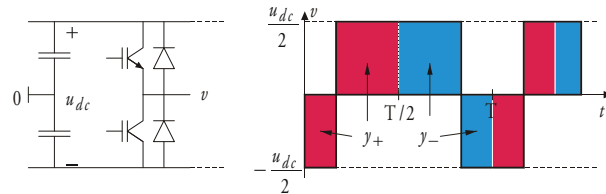


Figure 2.12: Modulation of a phase potential.

In this case the modulation is designed to control the position of both flanks. The maximum voltage time area for one half switching period is given by (2.18). Note that the output average voltage in this case can be both negative and positive with reference to the mid point in the DC link.

$$Y_0 = \frac{u_{dc}}{2} \cdot \frac{T}{2} \quad (2.18)$$

The voltage time area y can be controlled between Y_0 and $-Y_0$. According to Figure 2.12, the modulating wave can be written as

$$u_m(\tau_+, \tau_-) = \begin{cases} \frac{u_{dc}}{2} \left(1 - \frac{\tau_+}{T/4} \right) & 0 < \tau_+, \tau_- < T/2 \\ \frac{u_{dc}}{2} \left(\frac{\tau_-}{T/4} - 1 \right) & \end{cases} \quad (2.19)$$

This is a triangular wave varying between $\pm U_{dc}/2$. When the dc link voltage varies, the amplitude of the triangular carrier wave varies in proportion. The differences between the phase potentials create the output voltage of the converter. It can be expressed as:

$$u = v_a - v_b \quad (2.20)$$

The idea is to control the voltage time area of the output voltage (u) by controlling the voltage time area of the phase potentials (v_a, v_b). The pulse modulation can be done in several different ways, and it is therefore important to select the best possible. That method is shown in Figure 2.13.

Based on the reference value u^* for the average of the output voltage (u), two new reference values are calculated, one for each phase. To achieve an optimal control the calculation of the two final reference values should be done in the following way:

$$\begin{aligned} u_a^* &= \frac{u^*}{2} \\ u_b^* &= -\frac{u^*}{2} \end{aligned} \quad (2.21)$$

where u_a^* controls the phase potential v_a and u_b^* controls v_b .

By means of this way of calculating the phase voltage time area references, and by the fact that both phases are modulated by the same carrier wave, the output voltage will have a pulse frequency of twice the modulation frequency. The relatively high pulse frequency minimizes the current ripple and increases the number of control possibilities.

Two units build up the modulator for a 4-quadrant DC converter. Each unit controls its own phase potential, but the carrier wave generator is shared. Figure 2.13 shows the structure of the modulator.

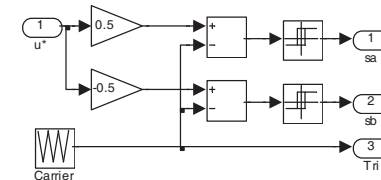


Figure 2.13: Modulator for a 4-quadrant DC converter

Figure 2.14 shows a four-quadrant DC-DC converter where the transistors are denoted T_i - T_o and the freewheeling diodes D_i - D_o .

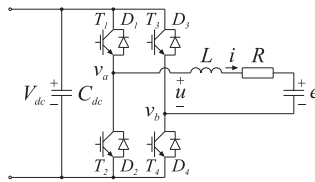


Figure 2.14: Four-quadrant DC-DC converter utilized in the study of modulation.

In the case of DC-DC conversion, the current controller inherently provides only a single voltage reference, the load voltage reference. However, for the converter in Figure 2.14, two transistor half bridges should be modulated. To solve this, the original load voltage reference is split in two phase potential references, one for each half bridge. Here, these phase potential references are denoted $v_{a, ref}$ and $v_{b, ref}$ which must be selected so that $v_{a, ref} - v_{b, ref} = v_{rf}$. These two references can be selected in any way as long as the difference between them equals the load voltage reference (in engineering units). Often, the phase potential references are chosen so that the risk of entering over-modulation is minimized. Over-modulation occurs when one or several of the phase potential references exceeds the peak of the carrier wave. A suitable choice is therefore to choose the phase potential references in such a way that two (both for the four-quadrant DC-DC converter) phase potential references reach over-modulation simultaneously for a certain load voltage reference. This means that the phase potential references should be selected according to

$$v_{a,ref} = \frac{v_{ref}}{2} \quad \text{and} \quad v_{b,ref} = -\frac{v_{ref}}{2} \quad (2.22)$$

This selection of modulator references where they are equal but with opposite sign, i.e. symmetrical with respect to the time axis, is referred to as symmetrical modulation. Overmodulation occurs when

$$v_{a,ref} = \hat{v}_{tri} \Leftrightarrow v_{b,ref} = -\hat{v}_{tri} \quad (2.23)$$

Since the maximum load voltage that can be obtained is equal to V_{ac} , this corresponds to the boundary between linear modulation and over-modulation, i.e.

$$\hat{v}_{tri} = v_{a,ref} = \frac{V_{dc}}{2} \quad (2.24)$$

The triangular carrier should thus have an amplitude of $V_d/2$. Usually, the current controller operates in engineering units (for example full scale of transducers often correspond to 10 V), which means that the triangular carrier must be scaled in the same way. Figure 2.15 shows typical modulator waveforms and converter output voltage and current, in the case of an ideal four-quadrant DC-DC converter.

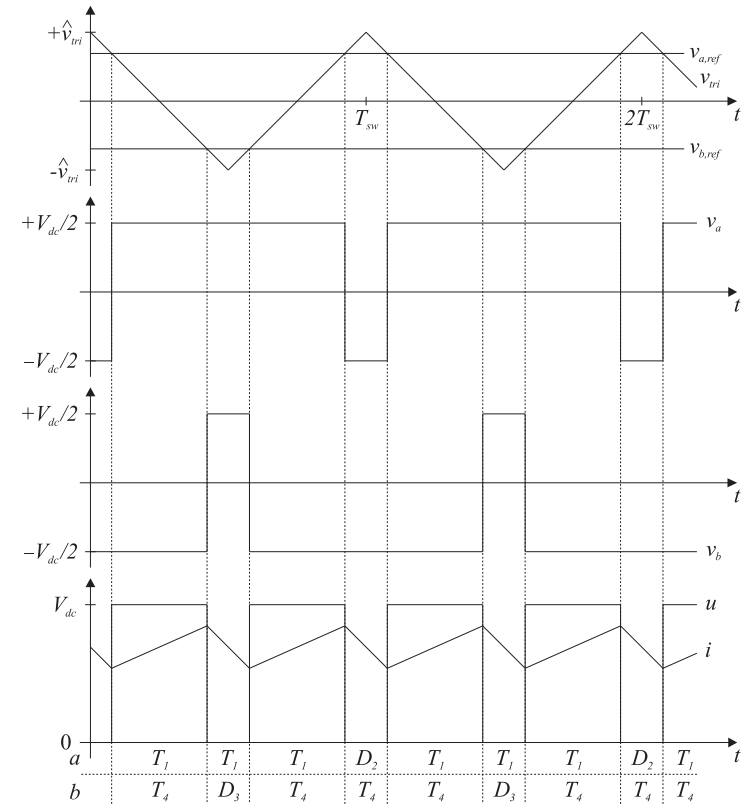


Figure 2.15: Modulation Waveforms and converter output voltage and current for an ideal four-quadrant DC-DC converter.

The output potentials of the half bridges are denoted v_a and v_b . The voltage across the load is denoted u , and is given by

$u = v_u - v_v$. The load current is denoted i . During one period of the triangular carrier, each half bridge output exhibits one positive and one negative step. The switching frequency is therefore equal to the frequency of the triangular carrier. If the period time of the triangular carrier is considerably shorter than the time constant of the load, the load current will exhibit a triangular shape as in Figure 2.15. In Figure 2.15 it is also shown which among the eight power semiconductors that are conducting in each sub-interval of the switching period. Note that the direction of the current play as important role in determining this.

As earlier stated, Figure 2.15 shows the modulator waveforms and the output voltage and current of the converter in the case of an ideal four-quadrant DC-DC converter. In reality,

power semiconductor devices cannot traverse from off-state to on-state and vice versa instantly. If both transistors of a half bridge operate in the on-state simultaneously there will be an extremely high power dissipation inside both since the DC link voltage will be divided between the two conducting devices but also since the current will be very high due to the low on-state resistance of the devices. To prevent this from occurring, during switching transients, transistor turn-on is delayed but not transistor turn-off. The duration of this time delay is termed blanking time or interlock time and the circuit performing the delay (which is a part of the transistor driver) is called blanking time or interlock circuit. Figure 2.16 shows the resulting modulation waveforms together with the output voltage and current of the four-quadrant DC-DC converter. Note that the average output voltage and current becomes lower than expected due to blanking time. Therefore, converter current controllers should be equipped with an integral part.

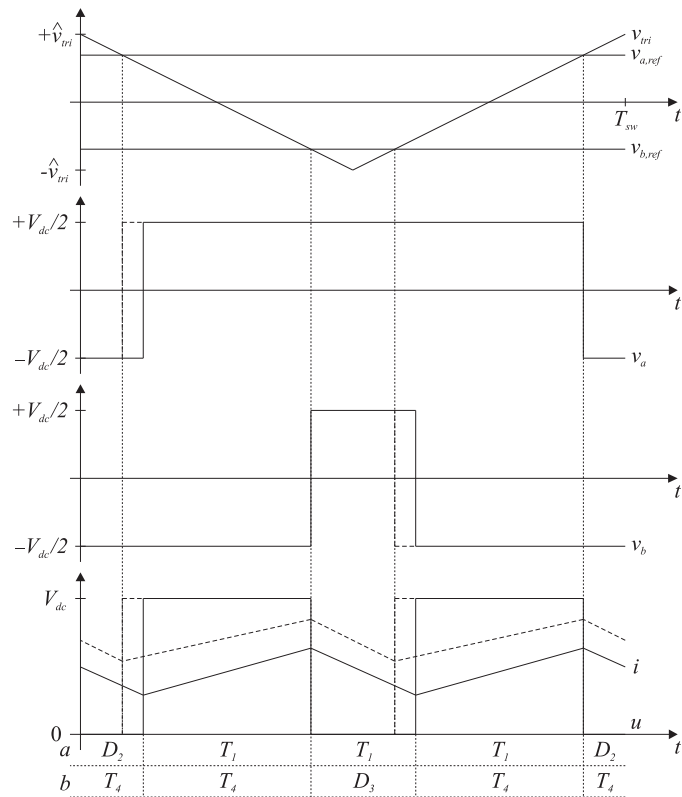


Figure 2.16: Modulation waveforms and converter output voltage and current for a four-quadrant DC-DC converter including blanking time effects. Note that the effect is strongly exaggerated.

This converter is called a DC converter, but as stated in Chapter 1 it can also be used for generation of a single-phase AC voltage, since the output voltage can have any polarity (Figure 2.17).

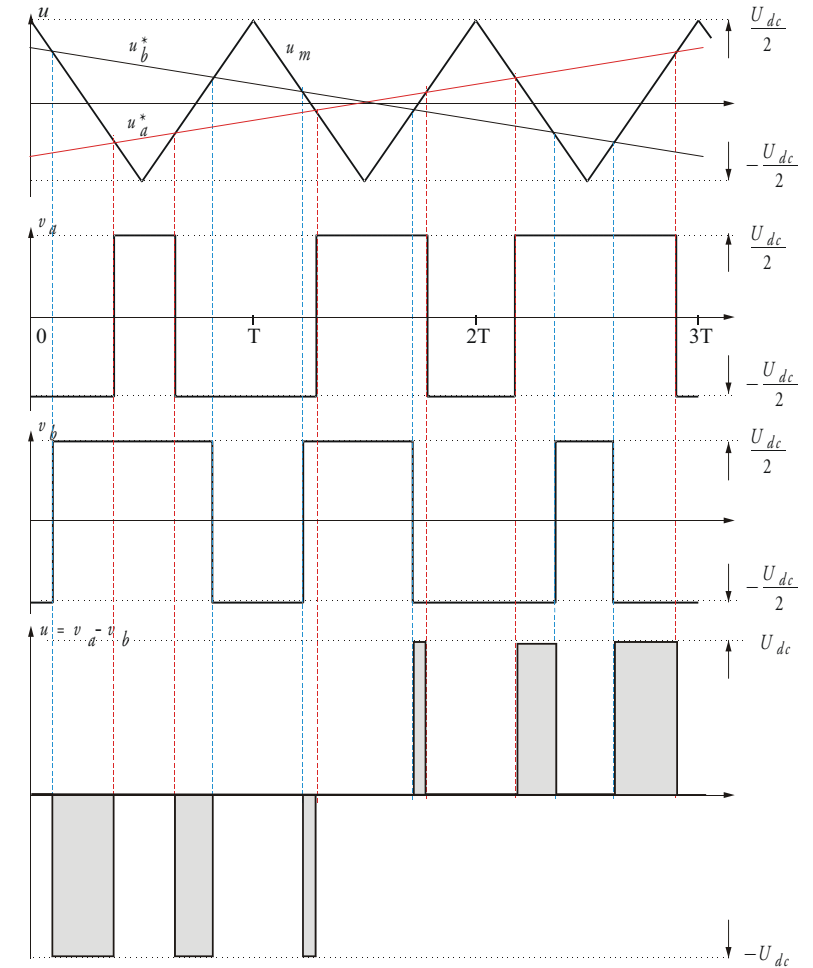


Figure 2.17: Example of modulation with 4 quadrant DC converter.

2.6 Modulation of three-phase converters

The three-phase converter is basically an extension of the 4-quadrant DC converter with one phase added to facilitate the generation of three individual phase potentials, used to control three-phase loads. The output voltage created can be supplied to a specific machine or constitute a three phase system, e.g. for uninterruptible power supplies (UPS). It is also possible to connect it to the utility grid (Figure 2.18). The three-phase converter consists of three transistor half-bridges in principle of operation corresponding to three switches s_a , s_b and s_c that are independently controlled.

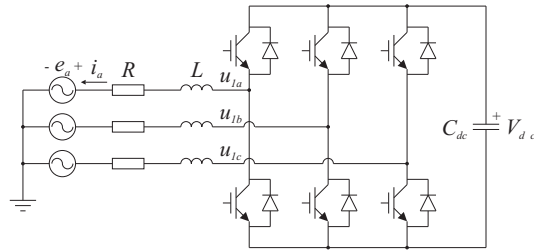


Figure 2.18: The investigated converter and grid connection.

Modulation schemes

The only difference between the investigated carrier wave modulation schemes for three-phase converters is the reference waveforms. To be more correct, the only difference is whether a zero-sequence signal is added or not, and if added, what the zero-sequence looks like [7]. A zero-sequence component in the references does not show up in the output voltage since the converter is not connected to the neutral. Therefore, the converter cannot produce any voltage affecting the neutral. A zero-sequence voltage per definition results in a zero-sequence current and since no neutral is connected there is no path for the zero-sequence (or sometimes called neutral) current to flow through. The converter investigated and its connection to the power grid, are shown in Figure 2.18.

This means that all the reference waveforms considered can be treated as being originally sinusoidal but later altered with a zero-sequence component. The reference waveforms are altered to affect the harmonic current spectrum or to increase the utilisation of the converter by increasing the maximum allowable modulation index for which linear modulation is obtained. The modulation index m_a is defined as

$$m_a = \frac{u_{i,ref}}{V_{dc}/2} \quad \text{where } i = a, b, c \quad (2.25)$$

where u is the converter output phase potential and V_{dc} the DC bus voltage. The reason why m_a is calculated using $V_{dc}/2$ is that in most cases the triangular carrier has a DC level equal to zero and amplitude proportional to $V_{dc}/2$. Here, the amplitude is equal to $V_{dc}/2$. This is because, otherwise, the reference signals should be scaled by the same factor as the

carrier peak-to-peak level in order to get an accurate modulation. Note that for full-bridge converters, the carrier DC level is not critical since the output voltages are differential, i.e. line-to-line voltages.

When m_a higher than 1 occurs, the switching of the particular phase is inhibited, since no crossings between the carrier and the reference waveforms occur. This situation is termed over-modulation. Over-modulation is equal to actuator saturation and should be avoided at least for stationary operation. The problem is not only that the controller command signals are not obeyed by the power electronic converter, but also that the output current harmonic content increases strongly [7][14].

The most important issue for all the investigated modulation methods is that the line-to-line output voltage should be equal for the different reference signals. As mentioned earlier, the voltage references are modified by a zero-sequence signal, in such a way that the maximum m_a is altered. The different voltage references are shown in Figure 2.19 for a time equal to one period of the fundamental.

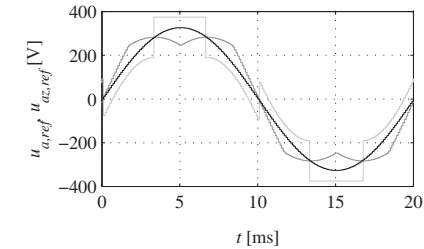


Figure 2.19: Carrier wave modulation: sinusoidal (black), symmetrical (grey) and bus-clamped (light-grey) voltage references.

The line-to-line output voltage resulting from these references equals 400 V, 50 Hz in all the cases. For sinusoidal modulation, the references are kept as they are, i.e. as sinusoidal phase potentials. For symmetrical references, the references are altered in such a way that the most positive and most negative reference is made equal but with opposite sign. For bus-clamped modulation the reference with the highest instantaneous value (positive or negative) is clamped to $\pm V_{dc}/2$, dependent on which is the closest.

For sinusoidal references it is clear that over-modulation, i.e. $m_a=1$, occurs when the reference peak equals half the DC link voltage. Therefore,

$$u_{i,ref,max} = \frac{V_{dc}}{2} \quad \text{where } i = a, b, c \quad (2.26)$$

for sinusoidal modulation. For symmetrical modulation the zero-sequence signal is calculated from

$$u_z = \frac{1}{2} \left(\max(u_{a,ref}, u_{b,ref}, u_{c,ref}) + \min(u_{a,ref}, u_{b,ref}, u_{c,ref}) \right) \quad (2.27)$$

For $1/6^{\text{th}}$ of the fundamental period where $u_{a,ref}$ is the most positive and $u_{b,ref}$ is the most negative reference in stationary conditions, the resulting reference for the half-bridge connected to the a -phase is written

$$u_{az,ref} = u_{a,ref} - u_z = \frac{\sqrt{3}}{2} \cdot \hat{u} \cdot \cos\left(\omega_1 t - \frac{\pi}{6}\right) \quad (2.28)$$

As in the previous case, over-modulation occurs when the reference equals half the DC link voltage, which gives

$$u_{iz,ref,max} = \frac{1}{\sqrt{3}} \cdot V_{dc} \quad \text{where } i = a, b, c \quad (2.29)$$

For bus-clamped modulation, the transition between two bus-clamped references is investigated. Assume that the b -phase has been clamped to $-V_{dc}/2$ and that the a -phase is going to be clamped to $+V_{dc}/2$. When this occurs

$$u_{a,ref} = \frac{\sqrt{3}}{2} \cdot \hat{u} \quad \text{and} \quad u_{b,ref} = -\frac{\sqrt{3}}{2} \cdot \hat{u} \quad (2.30)$$

This means that the zero-sequence voltage to be set equals

$$u_z = \frac{\sqrt{3}}{2} \cdot \hat{u} - \frac{V_{dc}}{2} < 0 \quad (2.31)$$

This gives

$$u_{az,ref} = \frac{\sqrt{3}}{2} \cdot \hat{u} - \left(\frac{\sqrt{3}}{2} \cdot \hat{u} - \frac{V_{dc}}{2} \right) = \frac{V_{dc}}{2} \quad (2.32)$$

and

$$u_{bz,ref} = -\frac{\sqrt{3}}{2} \cdot \hat{u} - \left(\frac{\sqrt{3}}{2} \cdot \hat{u} - \frac{V_{dc}}{2} \right) = \frac{V_{dc}}{2} - \sqrt{3} \cdot \hat{u} \quad (2.33)$$

The limit for over-modulation is given by

$$u_{bz,ref} = \frac{V_{dc}}{2} - \sqrt{3} \cdot \hat{u} \geq -\frac{V_{dc}}{2} \quad (2.34)$$

which gives

$$u_{iz,ref,max} = \frac{1}{\sqrt{3}} \cdot V_{dc} \quad \text{where } i = a, b, c \quad (2.35)$$

This means that modulation with symmetrical or bus-clamped references provides a 15 % higher margin to over-modulation compared to modulation with sinusoidal references. The harmonic content of the output voltage depends on the waveform of the reference voltage. Both the differential mode (DM) harmonics, i.e. the harmonic content in

$$u_{ij} = u_{iz} - u_{jz} \quad \text{where } i, j = a, b, c \quad (2.36)$$

and the common mode (CM) harmonic content, i.e. the harmonic content in u_o , are of great importance. For example, the DM harmonics result in losses in filter inductors etc. The CM harmonics can also result in parasitic current through capacitive coupling between conductors and ground. In turn, this results in malfunction of earth protectors and the occurrence of bearing currents in motors. The DM and CM harmonic content for the different modulation references in Figure 2.19 are shown in Figure 2.20 and Figure 2.21, respectively.

For a purely inductive filter, the damping is proportional to the frequency, since the impedance is given by

$$Z_L = j\omega_1 L \quad (2.37)$$

The resulting harmonic content in the output currents is shown in Figure 2.22, in the case of $L = L_{line} = 1.7$ mH. This filter inductance is suitable for a 100 kW converter with 750 V DC bus voltage connected to a 400 V three-phase AC grid [2]. Note that the harmonic current content is calculated from the DM spectrum of the output voltage.

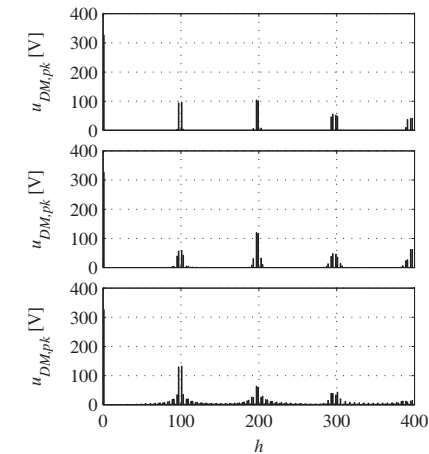


Figure 2.20: Differential mode output voltage spectrum for sinusoidal (*top*), symmetrical (*middle*) and bus-clamped modulation (*bottom*). The fundamental frequency is 50 Hz and the spectrums are shown as a function of harmonic order h , where 1 denotes the fundamental.

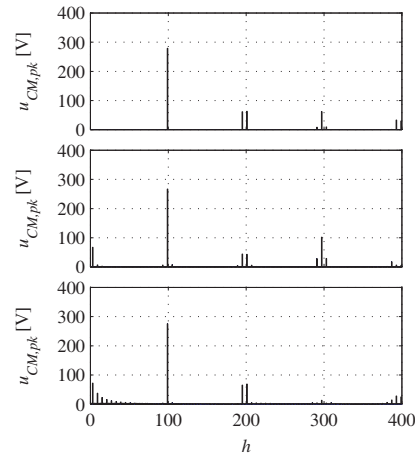


Figure 2.21: Common mode output voltage spectrum for sinusoidal (*top*), symmetrical (*middle*) and bus-clamped modulation (*bottom*). The fundamental frequency is 50 Hz and the spectrums are shown as a function of harmonic order h , where 1 denotes the fundamental.

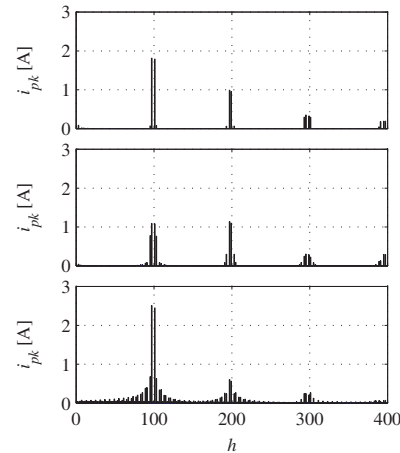


Figure 2.22: Output current spectrum for sinusoidal (*top*), symmetrical (*middle*) and bus-clamped modulation (*bottom*). The fundamental frequency is 50 Hz and the spectrums are shown as a function of harmonic order h , where 1 denotes the fundamental.

2.7 Vector representation

In the three phase self commutated converter with modulation of the phase potential in two levels ($\pm U_{dc}/2$) the switches can be combined in eight patterns. Each combination (s_a, s_b, s_c) is defined as a state. In Figure 2.23 are all combinations and corresponding states are shown. To every combination a set of phase voltages is related.

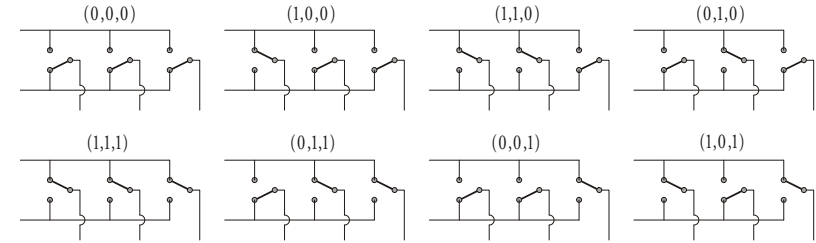


Figure 2.23: Eight combinations....

The combinations of phase voltages in Figure 2.23 can be expressed as complex space vectors. A space vector $\vec{x}^{\alpha\beta}$ is defined from instantaneous three-phase values x_a, x_b and x_c according to

$$\vec{x}^{\alpha\beta} = x_a \cdot e^{j0} + x_b \cdot e^{j\frac{2\pi}{3}} + x_c \cdot e^{j\frac{4\pi}{3}} = x_\alpha + jx_\beta \quad (2.38)$$

For the converter output voltage the expression above results in eight possible output voltage vectors. Two of the vectors are zero vectors and the other six create a star, see Figure 2.24.

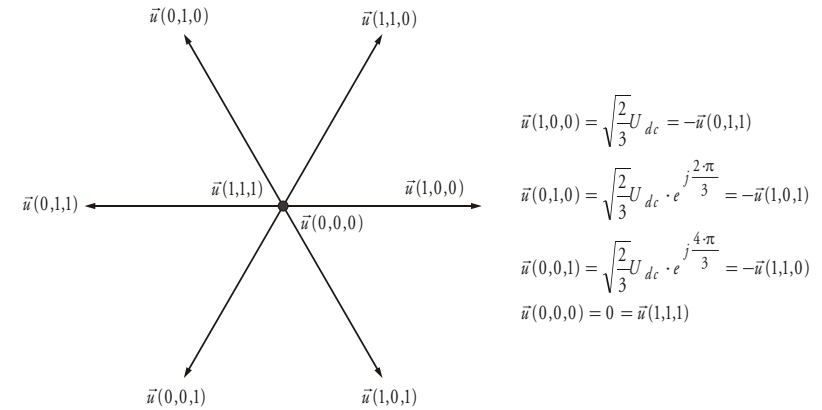


Figure 2.24: Voltage vectors from the 3-phase 2-level converter.

For general symmetrical three-phase quantities, i.e. if

$$\begin{cases} x_a = \hat{x} \cdot \cos(\omega_1 t) \\ x_b = \hat{x} \cdot \cos\left(\omega_1 t - \frac{2\pi}{3}\right) \\ x_c = \hat{x} \cdot \cos\left(\omega_1 t - \frac{4\pi}{3}\right) \end{cases} \quad (2.39)$$

the power-invariant transformation is given by

$$\begin{cases} x_\alpha = \sqrt{\frac{3}{2}} \cdot x_a \\ x_\beta = \frac{1}{\sqrt{2}}(x_b - x_c) \end{cases} \quad (2.40)$$

It is important to understand that the last expression cannot be used for switched quantities since these are not symmetrical. In the stationary $\alpha\beta$ -frame, the same calculation methods as used for regular three-phase calculations can be used. Kirchoff's voltage law in space coordinates applied to a three-phase converter with inductive filter connected to a three-phase grid (Figure 2.18) gives

$$\bar{u}^{\alpha\beta} - L \frac{d}{dt} \bar{i}^{\alpha\beta} - R \bar{i}^{\alpha\beta} - \bar{e}^{\alpha\beta} \approx \bar{u}^{\alpha\beta} - L \frac{d}{dt} \bar{i}^{\alpha\beta} - \bar{e}^{\alpha\beta} = 0 \quad (2.41)$$

This means that the current derivatives are expressed as

$$\begin{cases} \frac{di_\alpha}{dt} = \frac{1}{L}(u_\alpha - e_\alpha) \\ \frac{di_\beta}{dt} = \frac{1}{L}(u_\beta - e_\beta) \end{cases} \quad (2.42)$$

Assuming that the switching frequency period time T_{sw} is much longer than the time constant of the inductive filter, i.e. $T_{sw} = 1/f_{sw} \ll \tau_{el} = L/R$, implies that the current derivatives can be approximated according to

$$\begin{cases} \frac{\Delta i_\alpha}{\Delta t} = \frac{1}{L}(u_\alpha - e_\alpha) \\ \frac{\Delta i_\beta}{\Delta t} = \frac{1}{L}(u_\beta - e_\beta) \end{cases} \Leftrightarrow \begin{cases} \Delta i_\alpha = \frac{1}{L}(u_\alpha - e_\alpha)\Delta t \\ \Delta i_\beta = \frac{1}{L}(u_\beta - e_\beta)\Delta t \end{cases} \quad (2.43)$$

Since triangular carrier modulation is utilised, the duration of each interval can be calculated from their carrier crossings. For one half period (negative slope) of the carrier, the following expression is valid

$$u_{i,ref} = \frac{V_{dc}}{2} - \frac{V_{dc}}{T_s} \cdot t_i \Leftrightarrow t_i = \left(\frac{1}{2} - \frac{u_{i,ref}}{V_{dc}} \right) \cdot T_s \quad (2.44)$$

Note that Δt is the time duration for which a certain voltage vector is applied, in other words the distance between two consecutive carrier crossings for the different reference voltages. Actually, for the investigated carrier wave modulation strategies, the only difference is the time duration for which each different voltage vector is applied.

The sampling time T_s is selected as half the switching frequency time period, i.e. $T_s = T_{sw}/2$. Furthermore, the sampling instants are synchronised with the carrier in such a way that the sampling instants coincide with the carrier peaks. This is done in order to perform the samplings when the currents pass their average to avoid sampling distortion.

The grid is considered as being symmetrical, i.e.

$$\begin{cases} e_a = \hat{e} \cdot \cos(\omega_1 t) \\ e_b = \hat{e} \cdot \cos\left(\omega_1 t - \frac{2\pi}{3}\right) \\ e_c = \hat{e} \cdot \cos\left(\omega_1 t - \frac{4\pi}{3}\right) \end{cases} \quad (2.45)$$

The instantaneous grid voltages for $\omega_1 t = \pi/4$ are given by

$$\begin{cases} e_a = \frac{1}{\sqrt{2}} \hat{e} \\ e_b = \frac{\sqrt{3}-1}{2\sqrt{2}} \hat{e} \\ e_c = -\frac{\sqrt{3}+1}{2\sqrt{2}} \hat{e} \end{cases} \quad (2.46)$$

This gives

$$\begin{cases} e_\alpha = \sqrt{\frac{3}{2}} \cdot e_a = \frac{\sqrt{3}}{2} \cdot \hat{e} \\ e_\beta = \frac{1}{\sqrt{2}}(e_b - e_c) = \frac{\sqrt{3}}{2} \cdot \hat{e} \end{cases} \quad (2.47)$$

To calculate the converter output voltage (which is not a symmetrical three-phase quantity) equation (2.38) has to be used. The switches are considered as being ideal, i.e. the output voltage assumes the two discrete levels $\pm V_{dc}/2$. The voltage vectors applied for $\omega_1 t = \pi/4$ are shown in Figure 2.25.

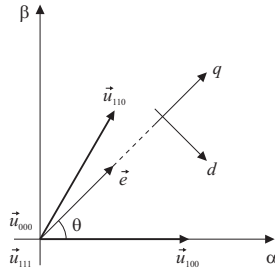


Figure 2.25: The grid voltage vector $\vec{e}$ and the converter output voltage vectors applied at $\omega_1 t = \pi/4$.

The voltage vectors are calculated in Table 2.1.

Table 2.1: Voltage vectors used at $\omega_1 t = \pi/4$.

Switch state	u_α	u_β
000	0	0
100	$\sqrt{2/3} \cdot V_{dc}$	0
110	$1/\sqrt{6} \cdot V_{dc}$	$1/\sqrt{2} \cdot V_{dc}$
111	0	0

In the following sections the switching behavior is investigated for the different modulation strategies. As in the previous section, the investigation is performed under the conditions listed in Table 2.2.

Table 2.2: Converter specification.

DC link voltage	V_{dc}	750 V
Grid peak voltage	$\hat{e}$	325 V
Grid frequency	f_1	50 Hz
Switching frequency	f_{sw}	5000 Hz
Line filter inductance	L	1.7 mH

Sinusoidal modulation

In the case of sinusoidal references in no-load condition, the stationary voltage reference for each phase is given by the instantaneous grid voltage. At $\omega_1 t = \pi/4$ this means that

$$\begin{cases} u_{a,ref} = \frac{1}{\sqrt{2}} \hat{e} = 229.8 \text{ V} \\ u_{b,ref} = \frac{\sqrt{3}-1}{2\sqrt{2}} \hat{e} = 84.1 \text{ V} \\ u_{c,ref} = -\frac{\sqrt{3}+1}{2\sqrt{2}} \hat{e} = -313.9 \text{ V} \end{cases} \quad (2.48)$$

The carrier and the converter output voltage references in the case of sinusoidal modulation are shown in Figure 2.26. Figure 2.26 also shows the switching states applied. Note that only the vectors depicted in Figure 2.25 are used, as expected.

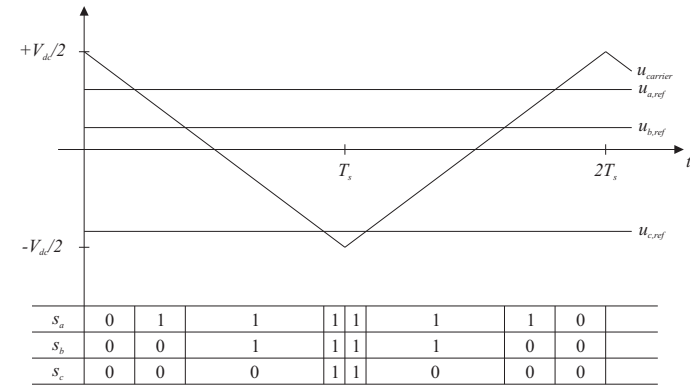


Figure 2.26: The modulation carrier and the voltage references at $\omega_1 t = \pi/4$ in the case of modulation with sinusoidal references. The lower part shows the applied switch states.

The corresponding carrier wave crossings occur at (calculated for the negative slope)

$$\begin{cases} t_a = \left(\frac{1}{2} - \frac{u_{a,ref}}{V_{dc}} \right) \cdot T_s = 0.194 \cdot T_s = 19.4 \mu\text{s} \\ t_b = \left(\frac{1}{2} - \frac{u_{b,ref}}{V_{dc}} \right) \cdot T_s = 0.388 \cdot T_s = 38.8 \mu\text{s} \\ t_c = \left(\frac{1}{2} - \frac{u_{c,ref}}{V_{dc}} \right) \cdot T_s = 0.919 \cdot T_s = 91.9 \mu\text{s} \end{cases} \quad (2.49)$$

The corresponding duration of each voltage vector (during the negative slope of the carrier) are, thus, equal to

$$\begin{cases} \Delta t_{000} = t_a - 0 = \left(\frac{1}{2} - \frac{u_{a,ref}}{V_{dc}} \right) \cdot T_s = 19.4 \mu s \\ \Delta t_{100} = t_b - t_a = \left(\frac{u_{a,ref}}{V_{dc}} - \frac{u_{b,ref}}{V_{dc}} \right) \cdot T_s = 19.4 \mu s \\ \Delta t_{110} = t_c - t_b = \left(\frac{u_{b,ref}}{V_{dc}} - \frac{u_{c,ref}}{V_{dc}} \right) \cdot T_s = 53.1 \mu s \\ \Delta t_{111} = T_s - t_c = \left(\frac{1}{2} + \frac{u_{c,ref}}{V_{dc}} \right) \cdot T_s = 8.1 \mu s \end{cases} \quad (2.50)$$

Applying this to equation (2.43) gives current vector variation according to Table 2.3.

Table 2.3: Current vector increase.

Switch state	Δi_α [A]	Δi_β [A]
000	-3.21	-3.21
100	3.78	-3.21
110	0.77	7.77
111	-1.34	-1.34

Note that the sum of the variations equals zero. This is approximately true if the pulse number m_f , i.e. the ratio between switching frequency and fundamental of the output voltage, is high. Also, if m_f is high, the grid voltage vector does not change much during the following carrier half period since the sampled grid voltages are almost unaltered. This yields that the same converter output voltage vectors appear in the opposite order for approximately the same duration as for the previous sampling interval. The space vector current trajectory for one switching interval is shown in Figure 2.27.

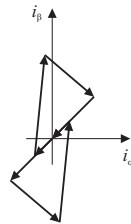


Figure 2.27: Current vectors for modulation with sinusoidal references at $\omega_1 t = \pi/4$.

Symmetrical modulation

For symmetrical references, in no-load operation, the stationary voltage reference for each phase is not given by the instantaneous grid voltage. Instead, the most positive reference

and the most negative reference are altered to have the same magnitude. This is done in the following manner and gives for $\omega_1 t = \pi/4$

$$\begin{aligned} u_z &= \frac{1}{2} \left(\max(u_{a,ref}, u_{b,ref}, u_{c,ref}) + \min(u_{a,ref}, u_{b,ref}, u_{c,ref}) \right) = \\ &= \frac{1}{2} (u_{a,ref} + u_{c,ref}) = \frac{1}{2} (229.8 - 313.9) \text{ V} = -42.05 \text{ V} \end{aligned} \quad (2.51)$$

This gives the voltage references

$$\begin{cases} u_{az,ref} = u_{a,ref} - u_z = 271.85 \text{ V} \\ u_{bz,ref} = u_{b,ref} - u_z = 126.15 \text{ V} \\ u_{cz,ref} = u_{c,ref} - u_z = -271.85 \text{ V} \end{cases} \quad (2.52)$$

The corresponding switching times (calculated for the negative slope of the carrier) are

$$\begin{cases} t_a = \left(\frac{1}{2} - \frac{u_{az,ref}}{V_{dc}} \right) \cdot T_s = 0.138 \cdot T_s = 13.8 \mu s \\ t_b = \left(\frac{1}{2} - \frac{u_{bz,ref}}{V_{dc}} \right) \cdot T_s = 0.332 \cdot T_s = 33.2 \mu s \\ t_c = \left(\frac{1}{2} - \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 0.863 \cdot T_s = 86.3 \mu s \end{cases} \quad (2.53)$$

The duration of each voltage vector (during the negative slope of the carrier) is thus

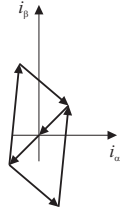
$$\begin{cases} \Delta t_{000} = t_a - 0 = \left(\frac{1}{2} - \frac{u_{az,ref}}{V_{dc}} \right) \cdot T_s = 13.8 \mu s \\ \Delta t_{100} = t_b - t_a = \left(\frac{u_{az,ref}}{V_{dc}} - \frac{u_{bz,ref}}{V_{dc}} \right) \cdot T_s = 19.4 \mu s \\ \Delta t_{110} = t_c - t_b = \left(\frac{u_{bz,ref}}{V_{dc}} - \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 53.1 \mu s \\ \Delta t_{111} = T_s - t_c = \left(\frac{1}{2} + \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 13.8 \mu s \end{cases} \quad (2.54)$$

Applying this to equation (2.43) gives current variation according to Table 2.4.

Table 2.4: Current vector increase.

Switch state	Δi_α [A]	Δi_β [A]
000	-2.28	-2.28
100	3.78	-3.21
110	0.77	7.77
111	-2.28	-2.28

Also in this case the sum of the variations equals zero. The space vector current trajectory for one switching interval, i.e. both the positive and the negative slope of the carrier, is shown in Figure 2.28.

**Figure 2.28:** Current vectors for modulation with symmetrical references at $\omega_1 t = \pi/4$.

Bus-clamping modulation

For bus-clamped references, in no-load condition, the stationary voltage reference for each phase is not given by the instantaneous grid voltage. Instead, the reference with the most positive or negative instantaneous voltage is clamped to a level equal plus or minus half the DC link voltage, i.e. $\pm V_{dc}/2$. As for symmetrical modulation, this is equal to calculating a neutral point voltage differing from zero. This is done in the following manner

$$u_z = \begin{cases} \max(u_{a,ref}, u_{b,ref}, u_{c,ref}) - \frac{V_{dc}}{2} & \text{if } \max(u_{i,ref}) > -\min(u_{i,ref}) \\ \min(u_{a,ref}, u_{b,ref}, u_{c,ref}) + \frac{V_{dc}}{2} & \text{if } \max(u_{i,ref}) < -\min(u_{i,ref}) \end{cases} \quad (2.55)$$

For $\omega_1 t = \pi/4$ this yields

$$u_z = \min(u_{a,ref}, u_{b,ref}, u_{c,ref}) + \frac{V_{dc}}{2} = u_{c,ref} + \frac{V_{dc}}{2} = 61.1 \text{ V} \quad (2.56)$$

The resulting voltage references are calculated according to

$$\begin{cases} u_{az,ref} = u_{a,ref} - u_z = 168.7 \text{ V} \\ u_{bz,ref} = u_{b,ref} - u_z = 23.0 \text{ V} \\ u_{cz,ref} = u_{c,ref} - u_z = -375.0 \text{ V} \end{cases} \quad (2.57)$$

The corresponding switching times (calculated for the negative slope of the carrier):

$$\begin{cases} t_a = \left(\frac{1}{2} - \frac{u_{az,ref}}{V_{dc}} \right) \cdot T_s = 0.275 \cdot T_s = 27.5 \mu\text{s} \\ t_b = \left(\frac{1}{2} - \frac{u_{bz,ref}}{V_{dc}} \right) \cdot T_s = 0.469 \cdot T_s = 46.9 \mu\text{s} \\ t_c = \left(\frac{1}{2} - \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 1.000 \cdot T_s = 100.0 \mu\text{s} \end{cases} \quad (2.58)$$

The corresponding duration of each voltage vector (during the negative slope of the carrier) is equal to

$$\begin{cases} \Delta t_{000} = t_a - 0 = \left(\frac{1}{2} - \frac{u_{az,ref}}{V_{dc}} \right) \cdot T_s = 27.5 \mu\text{s} \\ \Delta t_{100} = t_b - t_a = \left(\frac{u_{az,ref}}{V_{dc}} - \frac{u_{bz,ref}}{V_{dc}} \right) \cdot T_s = 19.4 \mu\text{s} \\ \Delta t_{110} = t_c - t_b = \left(\frac{u_{bz,ref}}{V_{dc}} - \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 53.1 \mu\text{s} \\ \Delta t_{111} = T_s - t_c = \left(\frac{1}{2} + \frac{u_{cz,ref}}{V_{dc}} \right) \cdot T_s = 0.0 \mu\text{s} \end{cases} \quad (2.59)$$

Applying this to equation (2.43) gives current increase according to Table 2.5.

Table 2.5: Current vector increase.

Switch state	Δi_α [A]	Δi_β [A]
000	-4.55	-4.55
100	3.78	-3.21
110	0.77	7.77
111	0.0	0.0

As before, the sum of the variations equals zero. The space vector current trajectory for one switching interval, i.e. both the positive and the negative slope of the carrier, is shown in Figure 2.29.

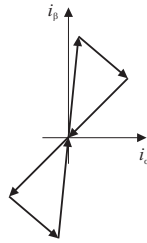


Figure 2.29: Current vectors for modulation with bus-clamped references at $\omega_1 t = \pi/4$.